Hussain Singapuri

AI ENGINEER (LLM, RAG, PYTHON)

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PROFESSIONAL SUMMARY

AI Engineer with 3 years of professional full-stack experience at a US-based medical-device company. Builds **LLM** applications and **AI agents** (Claude and Gemini APIs, tool use), **RAG** pipelines with embeddings, vector search and an eval harness, and **Python / FastAPI** REST services. Built **nanoserve**, an LLM inference engine in **PyTorch** with 3.1x throughput over sequential decoding. Uses **Docker**, **Git** and **GitHub Actions**; Claude Certified Developer; Bachelor's in Computer Science (AI & Machine Learning).

CORE COMPETENCIES

Generative AI & LLM Applications | RAG, Embeddings & Vector Databases (pgvector) | Agentic AI & Multi-agent Orchestration | LLM Evaluation & Prompt Engineering | Python, FastAPI & REST APIs | PyTorch & LLM Inference Serving | SQL & PostgreSQL | Docker, Git & CI/CD

WORK EXPERIENCE

NeuroNetworks AI Ed

2026 - Present

CTO & Principal Course Architect

Pune, India

- Built and run the website neuronetwrks.com with a Python static-site generator deployed on Vercel, including programmatic SEO
- Lead technology for an Indian AI and tech education brand (Neuronetworks AI Ed Private Limited) offering programmes for students after Class 10
- Architected the curriculum: a 1-year AI Foundation course and five 3-year specialisations (AI/ML Engineering, Software Development, Data Science & AI, Product & UX / AI Design, Cybersecurity)
- Designing Neuromind Practical AI Lab, a browser-based IDE and sandbox with an AI tutor: phased spec, server sizing, engineering team plan and CAPEX/OPEX model

US-based medical-device company

Mar 2023 - Sep 2026

Full Stack Developer

Remote

- Built and maintained full-stack web applications for the company
- Built **AI agents** for internal use
- Built several apps, including a **CRM**
- Performed penetration / security testing on 5+ of the company's mobile apps, all live on the Google Play Store

PROJECTS

nanoserve

Open Source

LLM inference engine built from scratch: paged KV cache, prefix caching, continuous batching with preemption, speculative decoding, OpenAI-compatible streaming API. 3.1x throughput over sequential decoding in benchmarks.

Python, PyTorch, FastAPI

site-to-agent

Open Source

Turns any website into a grounded chat agent: crawler, hybrid BM25 + vector retrieval with RRF, cited answers, SSRF-hardened fetching and an eval harness.

Next.js, TypeScript, pgvector

multi-agent-research-crew Open Source

Planner, parallel researcher, writer and verifier agents on a custom DAG orchestrator with retries, timeouts, token budgets and claim-level citation checking.

Python, asyncio, Claude

ai-lead-receptionist Open Source

AI receptionist for home-service businesses: missed-call text-back, SMS and web chat that qualifies leads and books jobs, owner dashboard.

Next.js, TypeScript, Twilio, Claude, Supabase

PageForge Portfolio project

Account-based marketing page generator: AI drafts only from approved facts and the account's public pages, a deterministic validator flags unsupported numbers, a reviewer signs off claim by claim, engagement flows to analytics and the CRM.

Next.js 16, React 19, Tailwind v4, Sanity, Supabase, Gemini, PostHog, HubSpot

Personal portfolio with full-stack backend Portfolio project

Lead pipeline with rate-limited enquiry form, booking with free intro calls and paid Stripe sessions (slot holds, signed webhooks), HMAC-signed CRM events, nonce-based CSP, 10 end-to-end API tests.

Vanilla ES modules, GSAP, Node 20, Postgres (Neon), Stripe, Resend

Orbital Surplus Portfolio project, fictional brand

Full-stack store with race-safe stock holds in one transaction, idempotent payment confirmation via signed Stripe webhook (test mode), customer accounts, admin with partial refunds; 9 end-to-end API tests.

three.js, GSAP, WebXR, Node 20, Postgres (Neon), Stripe, Resend

More projects

lead-intel-engine, contractor-ai-ops-dashboard, Nightjar & Co., Lumen, Kilnwood Roasters, Cairnmoor, Lowlight Video Club, Safey Inventory, TripSync

EDUCATION

Bachelor's, Computer Science (AI & Machine Learning) Vishwakarma Institute of Technology, Pune 2022-2025

Pune

CERTIFICATIONS

Claude Certified Developer	Anthropic, Foundations (CCDV-F)
Claude Certified Orchestrator	Anthropic, Foundations (CCAO-F)
Generative AI Leader	Google Cloud
Claude Code in Action	Anthropic
Claude Code 101	Anthropic

SKILLS

AI / LLM: Claude API, Gemini API, AI agents, multi-agent orchestration, tool use / function calling, RAG (hybrid BM25 + vector, RRF), evals, prompt-injection defence, LLM inference serving, Claude Code and AI coding agents

Languages: Python, SQL, TypeScript, JavaScript, Dart, HTML, CSS

Back end: Node.js, FastAPI, asyncio, REST APIs, webhooks (HMAC signing), auth (scrypt, sessions, throttling), Stripe Checkout and refunds, Resend

Data: PostgreSQL, Neon, Supabase (RLS, Edge Functions), PGlite, pgvector, SQLite, Sanity CMS

DevOps: Docker, Git, GitHub Actions, Vercel

Front end: React 19, Next.js, TypeScript, JavaScript, Tailwind CSS, shadcn/ui, TanStack Start / Query, GSAP, Lenis, three.js, WebGL, Canvas 2D, responsive and accessible UI, motion and interaction design

Security: mobile app penetration testing, CSP and security headers, SSRF hardening, input validation

Automation & integrations: workflow automation, CRM integrations (HubSpot), PostHog, Twilio SMS, web crawling / scraping

Mobile: Flutter, Riverpod, Firebase **Design:** UI/UX design, design systems, technical SEO, JSON-LD structured data, programmatic SEO